

Youngjin Yang

Undergraduate Researcher | Life Sciences
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Research Interest

My interests lie in biological systems engineering and bioprocess development, with an emphasis on designing, evaluating, and optimizing biological systems through careful experimentation and quantitative analysis. During 2026, I worked on RNA engineering research, developing experience in cell-based assays, molecular cloning, reporter-based screening, and experimental reproducibility.

Education

Chung-Ang University — 2026 – Present
B.S. in Life Sciences. GPA: 4.2 / 4.5 (current).

Dongguk University — Previous undergraduate study
Life Sciences; transferred to Chung-Ang University.

Research Experience

Chung-Ang University — Kim Hak-gyun Lab — 2026
Undergraduate Research Intern. Worked on circRNA translation optimization, including IRES screening and UTR optimization. Performed HEK293T cell culture, transient transfection, molecular cloning, dual-luciferase assays, and quantitative analysis. Screened a large IRES candidate set and evaluated reproducibility across repeated experiments.

Dongguk University — Im Gun-il Lab — Jun 2023 – Dec 2023
Undergraduate Research Intern, Regenerative Medicine Research Lab. Participated in cell-based regenerative medicine research, including FACS for evaluating cellular states in spheroid culture and bone-marrow preparation. Also participated in mouse and rabbit in vivo experiments evaluating therapeutic cell/drug-related effects in an osteoarthritis model, including GFP-based observation.

Selected Research Projects

circRNA Translation Optimization — Chung-Ang University · 2026
IRES screening for improving translation efficiency of circular RNA. Firefly/Gaussia dual-luciferase reporter system was used to compare candidates, with FHB confirmed as the WT normalization reference. Follow-up work examined combinations of selected IRES candidates with UTRs and considered repeatability and sequence/structural interactions.

IRES Screening & Reproducibility Analysis — 2026
Screened approximately 100 candidate conditions using HEK293T cells. Technical duplicates were measured after 24 h and normalized using Gaussia/Firefly ratios relative to WT (FHB). Repeated screening of the candidate set was used to assess reproducibility and select candidates for downstream optimization.

Technical Skills

Cell & Molecular Biology
HEK293T cell culture · transient transfection · Lipofectamine 3000 · dual-luciferase assay · FACS · PCR · molecular cloning · plasmid preparation

RNA & Biological Systems
circRNA engineering · IRES screening · UTR optimization · reporter assays · experimental design · reproducibility analysis

Data & Tools
Python · quantitative experimental data analysis · statistical analysis · data visualization · GitHub

Leadership & Activities

Student Council — Undergraduate
Participated in student council activities for two years and served as a department/student council leader.

External & Campus Projects — Undergraduate

Participated in Korea Social Contribution Association supporter activities and led university projects including curriculum development and creative convergence projects.

Training & Future Development

KAIST STAR Program — Planned

Targeting winter research internship opportunities to broaden experience in biological engineering and quantitative/data-driven research.

K-NIBRT Bioprocess Training — Planned

Planning biopharmaceutical process education with interest in upstream/bioprocessing applications.

Certifications & Language — Planned

TOEIC 900 target · TOEIC Speaking · TOEIC Writing preparation.

Career Direction

Long-term goal: Process Engineering in the biopharmaceutical industry, with particular interest in applying experimental optimization, cell-based systems, quantitative analysis, and biological engineering to scalable bioprocess development.